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Full Length Article

Electronic and optical properties of plutonium metal and oxides from Reflection Electron Energy Loss Spectroscopy

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ABSTRACT

Reflection Electron Energy Loss Spectra (REELS) have been acquired on plutonium δ-Pu-Ga alloys, α-Pu_2O_3 and PuO_2. The position of the peaks attributable to the Pu 6p excitation can be used as 'fingerprint' for these species. X-ray Photoelectron Spectroscopy has been used to aid interpretation of the REELS spectra. The band gap of PuO_2 was determined from the onset of the energy loss in the REELS spectrum. The REELS data have been evaluated within the dielectric response theory by means of the QUEELS-$\epsilon(k, w)$-REELS software [Surf. Interface Anal. 36, (2004) 824]. The Energy Loss Function, dielectric function, refractive index and extinction coefficient for these materials have been determined in the ultraviolet and extreme ultraviolet range and are compared to previously reported results. Crown copyright (C) 2021 Published by Elsevier B. V. All rights reserved.

1. Introduction

Plutonium metal and alloys are a hazardous material being both radioactive and toxic and requires safe storage for a prolonged period. In the pure form, plutonium metal exists in the monoclinic structure known as α-Pu at ambient conditions [1]. The high temperature face centered cubic phase, δ-Pu, can be stabilized at ambient conditions as a binary alloy with small concentrations of either aluminum, americium, cerium, gallium, or scandium [1]. Plutonium metal is electropositive and always covered with an oxide film, even in the purest inert atmosphere handling equipment. The oxide film consists of two crystallographically distinct structures of the bixbyite sesquioxide, α-Pu_2O_3 and fluorite PuO_2 as determined from X-ray Diffraction (XRD) measurements [2]. Similarly, two distinct electronic structures for the plutonium oxides have been detected by Ultraviolet and X-ray Photoelectron Spectroscopy (UPS, XPS) [3,4]. The corrosion of plutonium metal and alloys are influenced by the oxide composition, with a α-Pu_2O_3 film reported to be significantly more reactive than PuO_2 coating [5]. Therefore, to better understand and to be able to predict the behavior of plutonium metal and alloys in storage, knowledge of the fundamental properties, kinetics and reaction mechanisms are required. For example, the initial inspection of stored metal is normally undertaken visually and therefore any surface coloration will be indicative of the oxide film and composition which will be determined by the optical properties [1,5].

There is currently limited knowledge of the optical properties (refractive index and extinction coefficient) of plutonium metal and oxides. Here, experimental determination of the optical properties of these materials could help validate theoretical models from first principle calculations. Additionally, knowledge of the optical properties could be used to aid interpretation of experimental data where they are required as input parameters, as in the analysis of data acquired from optical methods. Optical properties are typically determined using laboratory based optical techniques such as reflectometry or ellipsometry at either a single or over a limited energy range. Larger energy ranges can be achieved using synchrotron techniques, however, the handling of such a hazardous material as plutonium at public synchrotron light sources is often prohibited or limited to a very small quantity. Furthermore, the surface oxide film may complicate the determination of the metal optical properties. An alternative approach is to use electrons to determine the Energy Loss Function (ELF) from Electron Energy Loss Spectroscopy (EELS). This has been performed in both transmission [6] and reflectance operating modes [7]. REELS, Reflection Electron Energy Loss Spectroscopy, can be acquired using a conventional Ultra High Vacuum electron spectrometer fitted with a suitable electron

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source. Here, a relatively low energy primary electron source (<5 keV) is used and the signal occurs from the surface (<5 nm depth) of the material. Methods to obtain the ELF and thus the optical properties of materials from the REELS data have been reported by multiple groups [8,9,10]. To date, none of these EELS techniques to determine the optical properties have been applied to the actinide elements. Indeed, there is only a single report of REELS measurements on plutonium which predated the methods to determine the optical properties [11]. In addition to determining the optical properties, information on electronic transitions [12] and the band gap [13] in semiconducting/insulating materials can be determined from REELS spectra.

REELS measurements are reported on two different δ-Pu-Ga alloys, α-Pu_2O_3 and PuO_2. The band gap for PuO_2 has been measured from the onset of the energy loss signal in the REELS spectrum. X-ray Photoelectron Spectroscopy has been used to aid interpretation of the REELS data. Finally, the optical properties of δ-Pu-Ga alloys, α-Pu_2O_3 and PuO_2 have been determined in the ultraviolet (UV) and extreme ultraviolet (EUV) ranges and are compared to the available data.

2. Experimental

REELS spectra were acquired using three different spectrometers operating with different geometries and resolutions as detailed below to acquire reproducible data on plutonium alloys/oxides. The object of the study was to acquire reproducible REELS data on plutonium alloys/oxides and not assess the operating performance of the individual spectrometers.

REELS spectra were acquired from a α-Pu_2O_3 film on a δ -Pu-Ga 3.2 at. % alloy substrate using an VG ESCALab MkII photoelectron spectrometer. Eight primary electron beam energies between 2020 and 622 eV were used to record the REELS spectra. The angle of incidence is 30° and reflectance of 0°. The resolution as defined by the FWHM (Full Width Half Maximum) of the elastic peak at 981 eV was 1.52 eV. XPS measurements were made using polychromatic Mg *K*α X-ray source and used to check the oxidation state and cleanliness of the sample prior to REELS analysis.

REELS spectra were acquired from 'as received' air oxidized PuO_2 film on δ-Pu-Ga 1.9 at. % metal substrate using a Physical Electronics PHI 710 Scanning Auger Nanoprobe. An area of the sample was sputter cleaned to obtain the spectra from the alloy substrate. Spectra were acquired using primary electron beam energies of 1 and 2 keV with an angle of incidence of 0° and reflectance of 42°. The resolution as defined by the FWHM of the elastic peak at 1 keV was 6.91 eV.

REELS spectra were acquired from a PuO_2 and α-Pu_2O_3 films on a δ-Pu-Ga 7 at. % alloy substrate using a Physical Electronics PHI Versaprobe III Scanning XPS microprobe with Auger. Primary electron energies of 1 and 2 keV were used with an incident angle of 55° and reflectance of 45°. Additional spectra were acquired on the PuO_2 film using an angle of reflectance of 0°. An area of the sample was sputter cleaned to obtain the spectra from the alloy substrate. The resolution as defined by the FWHM of the elastic peak at 1 keV was approximately 1.76 eV from average of eight spectra. XPS measurements were made using monochromatic Al *K*α 100 μm spot size X-ray source and used to check the oxidation state and cleanliness of the sample prior to REELS analysis.

3. Data analysis procedure

The data analysis procedure has been explained in detail elsewhere and only a brief overview will be provided here [12,14]. First, the multiple scattering contribution to the REELS spectrum are removed using the QUASES-XS-REELS (Quantitative Analysis of Surface Electron Spectra Cross Sections determined by REELS) software [15] following the method described by Tougarrd and Chorkendoff [16]. This affords the normalized experimental inelastic scattering cross section, K_{exp}. Secondly, the theoretical single-scattering cross section $K_{sc}(E, \hbar\omega, \theta_i, \theta_o)$ is determined (with E the energy of the incident electrons, ħω the energy loss, θ_i and θ_o the incident and reflected angles with respect to the sample surface normal) to compare to K_{exp}. This is implemented using the QUEELS-∊(*k*, ω)-REELS software (QUantitative analysis of Electron Energy Losses at Surfaces [17,18] where the input parameters are in the form of an ELF (Energy Loss Function), Im[-1/∊ (*k*, ω)] where ∊ (*k*, ω) is the dielectric function of the material. The ELF is represented by parametrized Drude-Lindhard oscillators [19]

$$\mathrm{Im}\left\{-\frac{1}{\epsilon(k,\omega)}\right\} = \sum_{i=1}^{n} \frac{A_i \hbar\gamma_i \hbar\omega}{\left(\hbar^2\omega_{0ik}^2 - \hbar^2\omega^2\right)^2 + \hbar^2\gamma_i^2\hbar^2\omega^2} \times \theta(\hbar\omega - E_g) f(\hbar\omega) \tag{1}$$

with the following dispersion relation,

$$\hbar\omega_{0ik} = \hbar\omega_{0i} + \alpha_i \frac{\hbar^2 k^2}{2m} \tag{2}$$

where A_i, ħγ_i, ħω_{0ik} and α_i are the strength, width, energy and dispersion of the *i*th oscillator, respectively. For semiconductors and insulators, the band gap E_g and a step function at the absorption edge θ(ħω-E_g) are included. In version 4.3 of the QUEELS-∊(*k*, ω)-REELS software the width and strength of the first oscillator can be modified with parameters α_1 and α_2 respectively to improve the accuracy of the determined ELF at low energy losses [12]. The oscillator strengths are adjusted to satisfy the optical sum rule

$$\frac{2}{\pi}\int_0^\infty Im\left\{-\frac{1}{\epsilon(\hbar,\omega)}\right\}\frac{d(\hbar\omega)}{\omega} = 1 - \frac{1}{n^2} \tag{3}$$

where n is the refractive index in the static limit. Here, the refractive index value of 2.1 for plutonium oxide thin films on a δ-Pu substrate reported by Larson *et al.* was used for analysis for both α-Pu_2O_3 and PuO_2 [20].

The minimum number of oscillators were used to obtain suitable ELF and corresponding K_{sc}. The mean oscillator parameters from the best fits to experimental data at different angles or primary electron energies are reported. Once a suitable agreement between the modelled K_{sc} and experimental K_{exp} was achieved the real part of the reciprocal of the complex dielectric function can be obtained along with the real and imaginary parts of dielectric functions ϵ_1 and ϵ_2 and refractive index, n and extinction coefficient κ from the ELF [21]. The acquired REELS spectra, comparisons between K_{exp} and K_{sc} and tabulated ELF, dielectric function and optical properties are in the supplementary information.

4. Results and discussion

To compare data acquired at different primary electron beam energies the energy scale of each REELS spectrum is referenced to the elastic peak set to zero eV. The experimental resolution, as determined by the Full Width Half Maximum of the elastic peak, was found to decrease with increasing primary beam energy on all three spectrometers. An example of the REELS data acquired from δ-Pu, α-Pu_2O_3 and PuO_2 using approximate 1 keV primary beam energy are shown in Fig. 1. The spectra in Fig. 1 are normalized on the height of the elastic peak. The energy loss spectra from all three materials are similar, consisting of three broad peaks between 8 and 45 eV. At low resolution two of these peaks merge to form one very broad peak. The peak positions increase in energy loss with expected oxidation state from plutonium metal to α-Pu_2O_3 and onto PuO_2. This increase in energy loss is similar to the binding energy increase observed in the Pu4f XPS spectra for these materials. Comparison of the REELS data acquired on the same material, e.g. alloy, α-Pu_2O_3 or PuO_2, but using different spectrometers showed identical energy loss positions for these broad peaks such that these could be used as a 'fingerprint' for the species similar to the peak position in Pu4f XPS. In addition to the three broad peaks, both oxides have shoulder on the low energy side of the first broad peak between 2 and 8

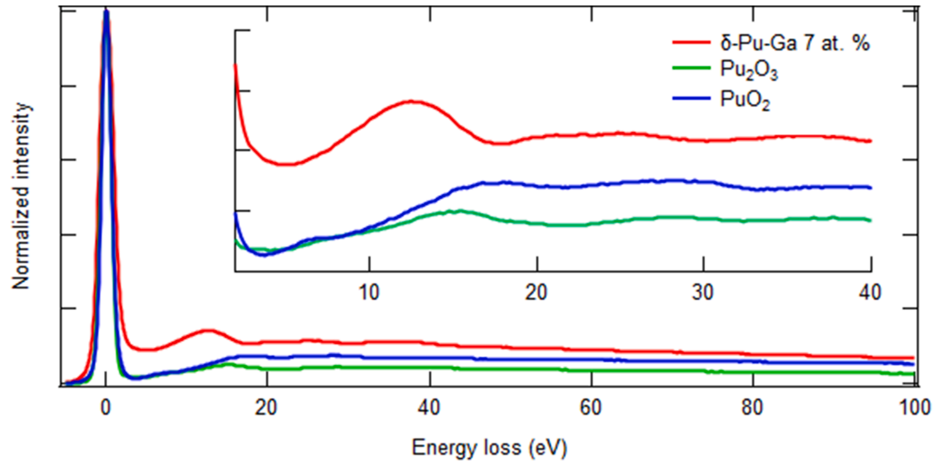

Fig. 1. Normalized REELS spectra of δ-Pu-Ga 7 at.% (red), α-Pu_2O_3 (green) and PuO_2 (blue). The low energy range just after the elastic peak and below 40 eV has been expanded highlighting the peak positions of the broad peaks of the three materials. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

eV. The origin of the peaks observed in the REELS spectra occur from surface and bulk plasmon excitations, intra-/inter-band transitions and core level ionisation processes as discussed below.

To determine if any surface plasmon excitations or other surface related phenomena could be detected, spectra were acquired with different primary electron energies and for PuO_2 using different sample emission angles with respect to the entrance of the electron analyser. The depth probed in the REELS spectra becomes less and the technique is more surface sensitive with lower primary electron energies or increased angle between the sample surface normal and electron analyser (angle of emission). There were no significant peak intensity changes in the REELS spectra on varying the sample angle or primary electron energy unlike those observed for aluminium [16]. The only changes in the spectra on changing primary beam energy were observed for δ-Pu and PuO_2. For δ-Pu this change consisted of the broad peak located between 20 and 30 eV becomes slightly more resolved at 1 keV into two possible broad peaks (Fig. 1). Whereas for PuO_2, the shoulder/peak between 4 and 8 eV disappeared on increasing the primary electron beam energy from 1 to 2 keV indicative of a surface plasmon peak. This is in agreement with the previous surface REELS study where the peak at 8 eV was reported to occur from surface plasmon and those at 14 and 27 eV from a bulk plasmon loss as measured on α-Pu sample with a residual oxygen and carbon impurities [11]. However, the decreased spectrometer resolution of the primary electron energy at 2 keV in this study may have obscured this low energy loss peak. Interestingly, the low energy peak was observed in the 2 keV REELS spectra after removal of the multiple scattering contributions and this may suggest that it is not from a surface plasmon excitation and more likely an inter-band transition (see below).

The possible core level excitations that can occur in 0 – 100 eV energy range are from the Pu $6p_{3/2}$, Pu $6p_{1/2}$, Pu 6 s and additionally for the oxides the O 2s. In the only other REELS study on plutonium these values were reported as 37, 53, and 75 eV for the Pu $6p_{3/2}$, Pu $6p_{1/2}$, and Pu 6s, respectively [11]. However, this study was undertaken prior to any XPS measurements on plutonium, which have subsequently shown, for α-Pu metal the core level binding energies occur at 17.2, 29.2/32, and 45.2/52.7 eV for Pu $6p_{3/2}$, Pu $6p_{1/2}$, and Pu 6s, respectively [22]. Clearly there is some discrepancy with the core level energy values reported from REELS and XPS measurements. Therefore, to aid interpretation of the REELS data reported here, valence band XPS measurements were acquired on the same sample on one of the spectrometers used in this study. The valence band XPS spectra for δ-Pu-Ga 7 at. %, α-Pu_2O_3 and PuO_2 are shown in Fig. 2 and in excellent agreement with those previously reported for plutonium metal [22,23] and oxides [24,25]. The spectra in Fig. 2 clearly show the Pu 6p spin-obit doublet occurs at 16.4 and 28.9 eV for δ-Pu and 17.8 and 30.4 eV for both α-Pu_2O_3 and PuO_2. The additional peak at 18.2 eV in the spectrum acquired on

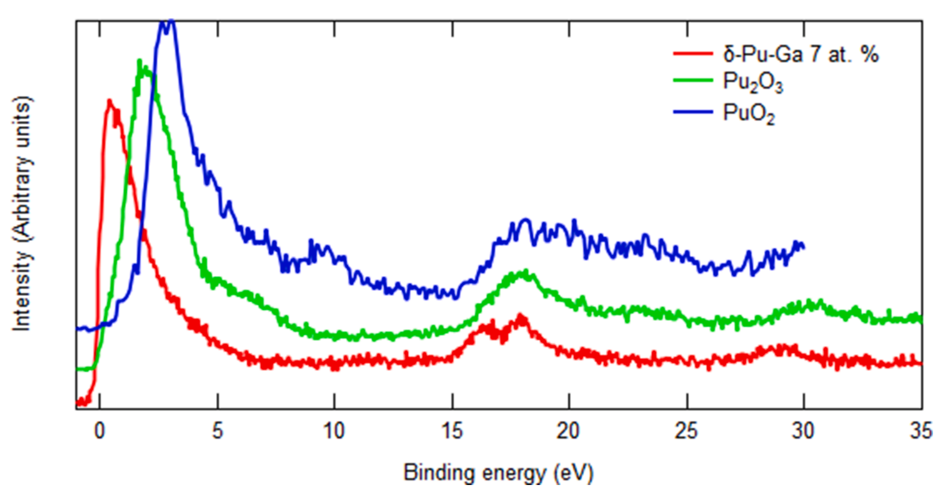

Fig. 2. Normalized XPS valence band spectra of δ-Pu-Ga 7 at.% (red), α-Pu_2O_3 (green) and PuO_2 (blue). Note the peak at 18.2 eV in the spectrum of the sputter cleaned alloy is attributable to the Ga 3d photoionization. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

sputter cleaned δ-Pu-Ga 7 at. % is attributable to the Ga 3d photoionization. It should be noted that the Ga 3d binding energy at 18.2 eV overlaps with the Pu $6p_{3/2}$ peak position of 17.8 eV in the valence band spectra of both oxides. However, we have not attributed any of this peak intensity to the presence of gallium in either of the oxides for two reasons. First, no other peaks attributable to Ga were detected in the survey spectrum of the oxides and secondly, the spectra in Fig. 2 are identical to spectra acquired from the oxide thin films formed on pure, non-alloyed plutonium [24,25]. The O 2 s photoionization peak occurs at 23 eV in both oxides. Comparison of XPS spectra in Fig. 2 to the REELS spectra in Fig. 1 strongly suggests that the broad peaks in the REELS spectra between 10 and 20 eV occur from Pu $6p_{3/2}$ core ionization and the broad peak between 20 and 30 eV is attributable to the Pu $6p_{1/2}$ core level ionization processes. Indeed, this is achieved by overlaying the corresponding XPS on the REELS spectrum for each of δ-Pu, α-Pu_2O_3 and PuO_2, and aligning the Pu 6p XPS to the REELS peaks by shifting the abscissa by approximately 4, 3, and 1.5 eV, respectively. The Pu 6p peaks in the XPS spectra occur at higher binding energies compared to the energy loss values in the REELS data due to the different final states between these techniques [26]. Finally, taking into account the peak position information described above suggests the surface species examined in the previous REELS analysis is PuO_2 [11].

One interesting observation in Fig. 2 is the $6p_{3/2}$ peak for both oxides occur at the same binding energy in the XPS spectra. It is worth noting that the XPS valence band is clearly shifted with increasing oxidation state and has also changed shape, as have the peaks attributable to Pu 4f spin-orbit doublet in both oxides (not shown). However, an increase in energy loss Pu $6p_{3/2}$ peak position is observed from the trivalent α-Pu_2O_3 to tetravalent PuO_2 in the REELS spectra (Fig. 1). One possible explanation is the different ionization processes and final states between XPS and REELS techniques, with REELS affording some indicative guide to the energy difference in conduction band position. This does not account why the increase in binding energy from α-Pu_2O_3 to PuO_2 is observed in the XPS Pu 4f peak positions but not the Pu $6p_{3/2}$ core level. One possible explanation may be attributed to $6p_{3/2}$ hybridization with the valence band as reported for uranium metal [27].

As stated in the introduction a reasonable estimate of the band gap can be determined from the onset of the energy loss in the REELS spectrum [13]. This assumes that the material is homogenous within the depth probed by the electron beam. This technique is limited to materials with band gap energies greater than 2.5–3 eV [13]. This limitation arises from the resolution or FWHM of the elastic peak which can overlap with the onset of the energy loss at lower energies. α-Pu_2O_3 has a reported band gap of 0.7 eV as determined from a combination of UPS and Inverse Photoemission Spectroscopy (IPES) and therefore cannot be measured from the REELS spectra [28]. A band gap of 3.92 eV was determined for PuO_2 from the REELS spectra acquired using a 1 keV primary electron energy. The increased FWHM (decreased resolution) of the elastic peak in the REELS acquired using a 2 keV primary beam energy precluded determination of suitable values of the band gap for PuO_2. The band gap value of 3.92 eV measured here is very similar to 4.1 eV determined from combined UPS-IPES measurements [28].

Fig. 3 shows the comparison between K_{exp} and K_{sc} and the corresponding ELF obtained for the two different δ-Pu-Ga alloys at 1 keV primary electron energy. The oscillator parameters used to model the ELF for both alloys are detailed in Table 1 along with possible assignments. The main difference in the ELF between the two alloys is in the energy of the first two oscillators and the intensity distribution in the 15–25 eV range. This clearly shows the ELF is weighted to higher energy for the δ-Pu-Ga 1.9 at. % alloy. However, it should be noted the spectra

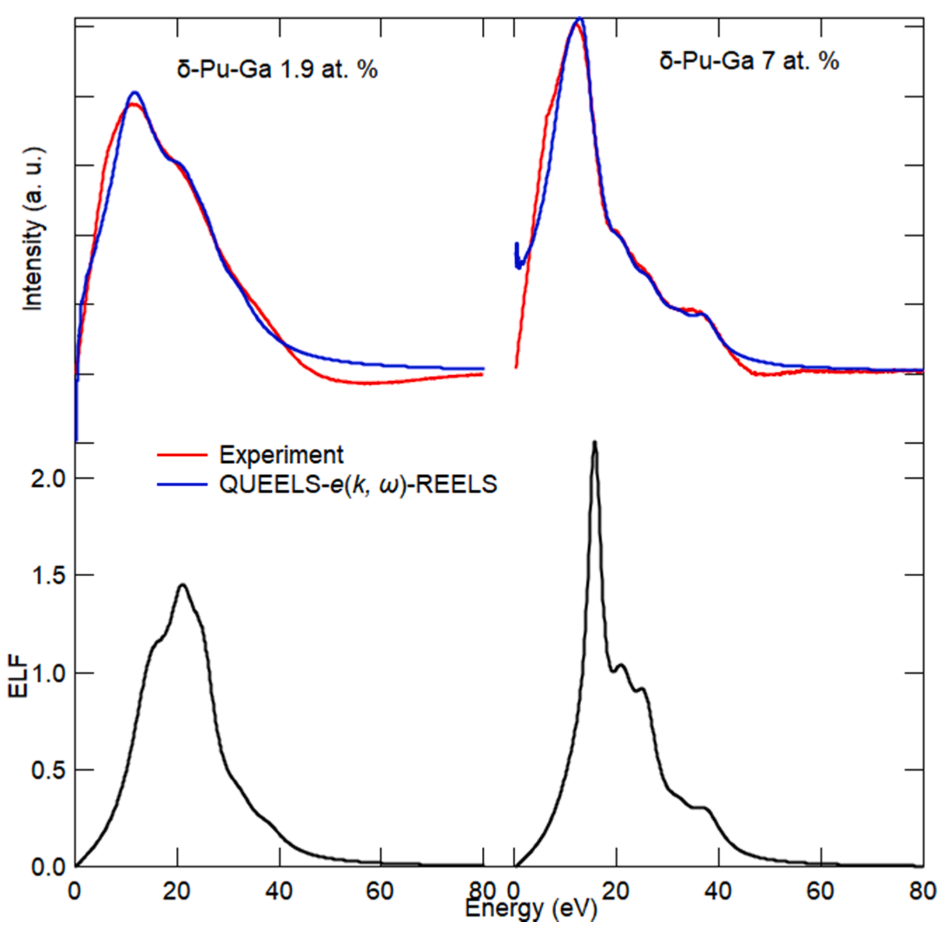

Fig. 3. Comparison between the K_{exp} and K_{sc} (top) and the ELF used to model K_{sc} (bottom) for the two δ-Pu-Ga alloys 1.9 at. % (left) and 7 at. % (right).

Table 1
Parameters used to model the ELF of δ-Pu-Ga alloys.

Alloy	i	$\hbar\omega_{0i}$ (eV)	A_i (eV^2)	$\hbar\gamma_i$ (eV)	α	Assignment
δ-Pu-Ga 1.9 at. %, E_g = 0	1	14.2	25.59	10	1	Surface plasmon
	2	15.9	93.73	9.15	1	Pu $6p_{3/2}$
	3	21.2	132.79	7	1	Bulk plasmon
	4	25.2	97.03	6	1	Pu $6p_{1/2}$
	5	32.3	39.84	7.8	0	
	6	38.1	23.67	7.5	0	
δ-Pu-Ga 7 at. %, E_g = 0	1	13.1	44.64	10	1	Surface plasmon
	2	15.65	89.67	3.3	1	Pu $6p_{3/2}$
	3	21.2	80.50	6.15	1	Bulk plasmon
	4	25.6	84.21	5.75	1	Pu $6p_{1/2}$
	5	32.3	30.92	6.9	0	
	6	37.7	49.49	7	0	

of the two alloys were acquired using different spectrometer resolutions and operating geometries. Indeed, it is likely that the main difference between K_{exp} and the ELF for the two alloys arises from the spectrometer resolution, which is evident in the oscillator widths, $\hbar\gamma_i$, which are smaller for the data acquired at higher resolution. The oscillator positions assigned to the core Pu 6p levels were from comparison to the XPS spectrum in Fig. 2. The oscillator at 21.2 eV in both alloys is attributed to a bulk plasmon excitation, which is expected at 20.9 eV for this material as calculated using the free electron plasmon energy equation, $E_p = 28.8 (N_v\rho/M)^{1/2}$, where N_v is the number of valence electrons per atom, ρ is the density and M is the atomic weight [29]. The first, low energy oscillator position could be tentatively ascribed to a surface plasmon as the calculated value from the bulk plasmon is expected to be between 12.1 and 14.8 eV [19].

The refractive index and extinction coefficient are shown in Fig. 4 for the 0 to 80 eV energy range for both plutonium alloys. An expanded plot for energies below 5 eV are shown in Fig. 5 along with the calculated optical properties from the Drude free electron theory [21] using the room temperature resistivity values for Pu-Ga alloys reported by Roux *et al.* [30] and experimental optical property values determined from single and spectroscopic ellipsometry measurements [31–33]. The optical properties determined here follow the shape and trend as those calculated and determined by ellipsometry. The refractive index and extinction coefficient for both alloys determined here are almost identical, deviating slightly in the 15–20 eV range, the same as the differences in the ELF (Fig. 3). This suggests alloying concentration has an insignificant effect on the optical properties. The optical properties calculated using Drude free electron theory using the room temperature resistivity values for gallium alloy concentrations between 3 and 10 at. % are essentially identical, adding confidence in the values determined from the REELS measurements here, and that any observed differences between the alloys most likely occur from the different spectrometer resolutions. The optical properties determined by REELS are lower than those calculated using the Drude free electron theory and higher than those derived from ellipsometry measurements of an alloy with a thin oxide film [31]. The optical property values reported here are comparable with those determined from ellipsometry measurements on sputter cleaned plutonium [20] and extracted from a multi-sample analysis combined with scanning electron microscopy [33]. The slightly lower extinction coefficient values determined here are indicative of decreased density [34], possibly from sputter damage or voids induced in the cleaning process for both these alloy samples. The preferential sputter removal of Ga has been reported for Pu-Ga alloys [35]. Although sputtering was used to afford a clean alloy sample in the single wavelength ellipsometry study [20], the depth probed in ellipsometry is greater than the REELS experiment so any sputter induced damage will alter the optical properties more in the latter technique. Any changes in the extinction coefficient will also be observed in the refractive index as both properties are related via Kramers-Kronig relationship and are calculated from the ELF here [18,21].

REELS spectra were acquired on α-Pu_2O_3 surface films on δ-Pu-Ga alloy substrates using two different spectrometers. The spectra were identical at loss energies greater than 8 eV, however, spectra acquired using higher resolution afforded the low energy loss peak at 4 eV, which was present in all spectra measured using primary electron beam energies from 620 to 2020 eV. Only the higher resolution data will be considered here. The comparison between K_{exp} and K_{sc} for α-Pu_2O_3 using an approximately 1 keV primary electron energy and the ELF are shown in Fig. 6. The parameters used to model the ELF are listed in Table 2 along with the possible assignments. As previously discussed, the Pu 6p positions were determined from comparison with the XPS spectra. There are no oscillator positions that occur at a similar energy to the O 2s of 23 eV or free electron bulk plasmon of 23.4 eV. The low energy oscillator position of 4 eV most likely occurs from an inter-band transition. A similar energy of 4.4 eV has been reported for the XPS shake up satellites for this oxide. XPS shake up satellites can occur from the energy loss of an outgoing photoelectron to promote a valence electron into the conduction band. From the combined UPS and IPES spectroscopic results this suggests that this occurs from the promotion of O 2p electron to the unoccupied 5f states [28]. Interestingly, additional XPS shake up satellites were reported for α-Pu_2O_3 at 16 eV, however, the authors were unclear of the origin of these peaks [36]. It can be seen from aligning the elastic peak of the REELS to the respective Pu 4f XPS peaks that these shake up satellites match the REELS Pu $6p_{3/2}$ ionization (see Figure in supplementary information) and therefore can be attributed to an intrinsic energy loss in the XPS spectrum of α-Pu_2O_3.

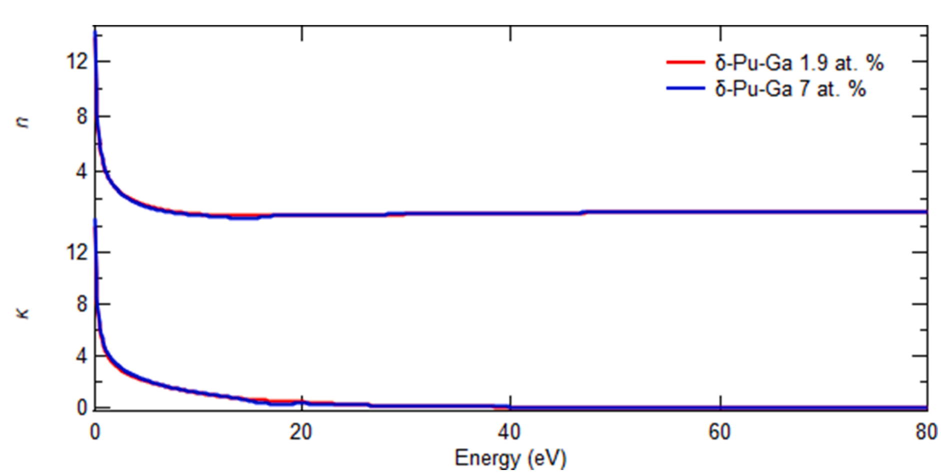

Fig. 4. Optical properties, n and κ, for two different δ-Pu-Ga alloys determined from REELS acquired using different spectrometers.

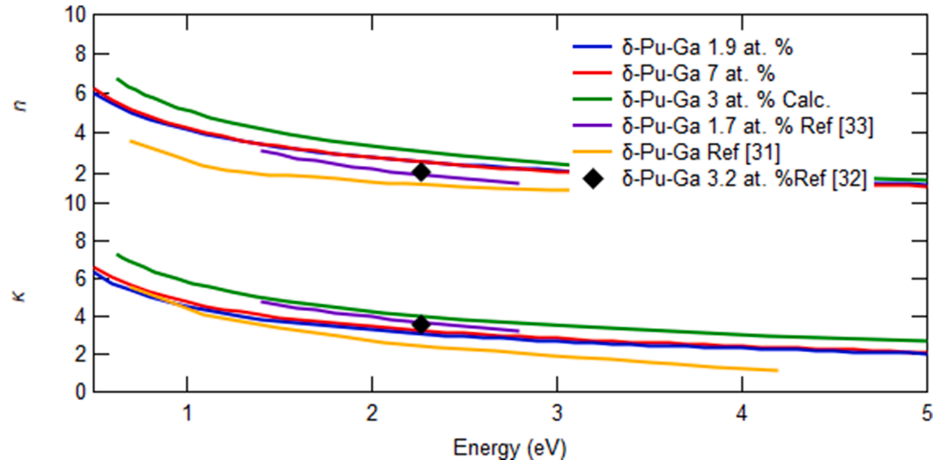

Fig. 5. Expanded region of Fig. 4 comparing the refractive index and extinction coefficient results from REELS measurements to those determined from ellipsometry and calculated using the Drude free electron model from electrical resistivity.

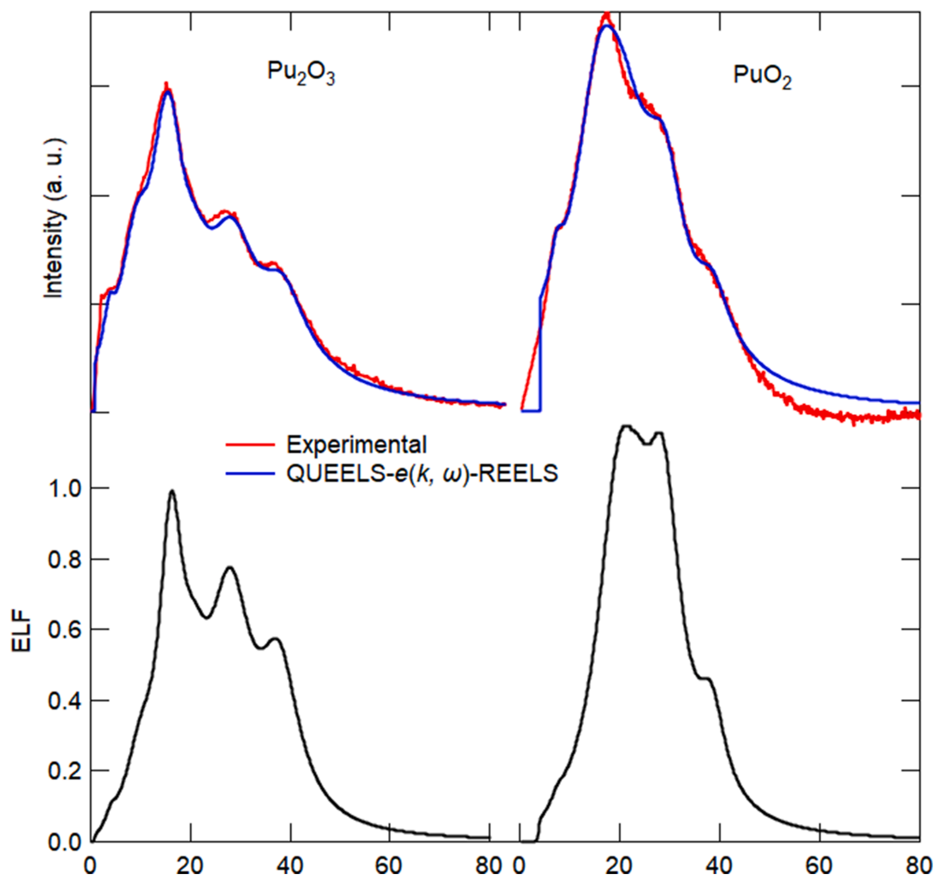

Fig. 6. Comparison between the K_{exp} and K_{sc} for α-Pu_2O_3 (top left) and PuO_2 (top right) and the ELF for α-Pu_2O_3 (bottom left) and PuO_2 (bottom right).

Table 2
Oscillator parameters used to model the ELF of α-Pu_2O_3.

Oxide	i	$\hbar\omega_{0i}$ (eV)	A_i (eV^2)	$\hbar\gamma_i$ (eV)	α	Assignment
α-Pu_2O_3, $E_g = 0.7$ eV,n = 2.1	1	4	0.23	2	0.2	O 2p → Pu 5f
	2	10.3	12.63	7	0.2	
	3	16.1	70.39	5.75	0.2	Pu $6p_{3/2}$
	4	21	33.14	7	0.2	
	5	28	152.81	10.25	0.2	Pu $6p_{1/2}$
	6	38	154.37	11.25	0.2	

Similar to α-Pu_2O_3, REELS spectra were acquired on PuO_2 surface films using two different spectrometers operating at different resolutions. Suitable removal of the elastic peak and multiple scattering contributions could not be achieved with the lower resolution data and this was not considered further. Comparison between K_{exp} and K_{sc} and the ELF are shown in Fig. 6. The parameters used to model the ELF are listed in Table 3 along with possible assignments. Here, we follow the recommendation by Pauly *et al.* [12] where the software values $E_g + \alpha_1$ are set to the band gap value of 3.92 eV determined from the REELS spectra and the software setting of E_g is set to the optical gap of 2.8 eV [37]. The oscillators attributable to the Pu 6p core levels were identified

Table 3
Oscillator parameters used to model the ELF of PuO_2.

Oxide	i	$\hbar\omega_{0i}$ (eV)	A_i (eV^2)	$\hbar\gamma_i$ (eV)	α	Assignment
PuO_2,	1	7.5	0.61	3	0.2	O 2p → Pu 5f
$E_g = 2.8$ eV,	2	20.38	192.93	10.5	0.2	Pu $6p_{3/2}$
$\alpha_1 = 1.12$,	3	23.9	26.5	6	0.2	
$\alpha_2 = 1$,	4	28.8	206	8.7	0.2	Pu $6p_{1/2}$
n = 2.1	5	39.1	60.89	6.56	0.2	

by comparison to the XPS spectrum. The oscillator located at 23.9 eV could either occur from the O 2s ionization at 23 eV or from a bulk plasmon excitation calculated to be 26.5 eV. Like the analysis of REELS data from α-Pu_2O_3, the oscillator at 7.5 eV most likely occurs from an inter-band transition. Here, as described above we turn to the photoemission data for the interpretation. The Pu 4f XPS spectra of PuO_2 shows distinctive shake up satellites 7 eV to lower kinetic energy to the main elastic peak [4,24,25]. These shake up satellites have been attributed to the O 2p occupied to Pu 5f unoccupied transition [28] and this is the most likely interpretation here.

The optical properties, n and κ in the energy range of 0 to 80 eV for both α-Pu_2O_3 and PuO_2 are shown in Fig. 7. Fig. 8 shows the data from Fig. 7 on an expanded abscissa including the two previously reported values of optical properties determined from ellipsometry measurements. The optical properties for PuO_2 found in this work are in excellent agreement with the values determined from spectroscopic ellipsometry measurements on single crystalline thin films reported by McClesky *et al.* [37]. There are no experimentally determined optical property values of for α-Pu_2O_3 to compare to, however, Larson *et al.* reported values for the optical properties of a plutonium oxide film which are slightly lower and higher than the respective refractive index and extinction coefficient of α-Pu_2O_3 determined here [20]. Indeed, the refractive index for plutonium oxide reported by Larson *et al.* was used in Eq. (3) for the analysis of both oxides here. Using lower values of the refractive index in the optical sum rule (Eq. (3)) for the analysis of α-Pu_2O_3 data afforded a slightly closer agreement with the optical properties for plutonium oxide reported by Larson *et al.* but still not a good match and this requires further discussion on these optical properties below. Finally, there are two reports of first-principles calculations on both α-Pu_2O_3 and PuO_2 to derive either the reflectivity [38] or the ELF, dielectric constants and refractive index [39], however, these do not compare well with the values determined here most likely due to the theoretical assumptions within the calculations like the value of Coulomb energy parameter, U, which effects the calculated electronic structure, band gap and hence the optical properties for the two oxides.

Comparison to the optical properties determined here to those reported by Larson *et al.* necessitates further discussion. The single wavelength values of the refractive index and extinction coefficient for plutonium oxide reported by Larson *et al.* were determined from growing oxide film at 60 °C [20]. As discussed in the introduction, the oxidation of plutonium metal forms two distinct oxide phases as characterized using XRD, UPS, XPS, and in the REELS data presented here. Realistically, it is likely that an oxygen concentration gradient exists, decreasing from the gas-oxide to the oxide-metal interface similar to that reported for zirconium oxidation [40]. This stoichiometry change has been observed as the expansion in oxide lattice parameters in diffraction data [41,42] and decreasing Pu:O ratio in XPS [36,43]. Nevertheless, the single values of n and κ reported for plutonium oxide reported by Larson *et al.* most likely represent a composite of the two oxides. The optical properties of a composite or homogeneous mixture of materials is the sum of the fraction of each species present [34] and using the optical properties determined here, it suggests the plutonium oxide film reported by Larson *et al.* has the composition of α-Pu_2O_3 97.6%, Pu metal 2.3% and PuO_2 0.1%. This assumes the composite consists of a homogenous mixture, which is incorrect, and a more rigorous multilayer Effective Medium Approximation model such as those employed in ellipsometry data analysis should be applied [40]. Although that type of analysis is outside the scope of the work undertaken here, it is worth noting that the oxide composition predominantly consists of α-Pu_2O_3 in the study reported by Larson *et al.* [20]. This is in excellent agreement with XPS analysis of oxide films formed following identical oxidation conditions [44]. Larson *et al.* studied the low pressure and low temperature oxidation (6.66 Pa, <100 °C) [20], whereas Stakebake has reported the oxidation of plutonium-gallium alloys at higher pressures and temperatures above 150 °C and did not detect the α-Pu_2O_3 phase [45]. This would suggest the oxide composition of films formed on plutonium metal and alloys stored at ambient conditions (for example room temperature and dry nitrogen gas) will predominantly consist of α-Pu_2O_3 with a small surface quantity of PuO_2.

5. Conclusions

Reflection Electron Energy Loss Spectra have been acquired on fcc plutonium-gallium alloys and plutonium oxides, α-Pu_2O_3 and PuO_2. X-ray Photoelectron Spectroscopy was used to aid assignment of the broad loss peaks in the REELS spectra. Within this limited study, it was found that the position of the Pu $6p_{3/2}$ core level could be used as a fingerprint for these species. From the onset of the energy loss signal in the REELS spectra the band gap of PuO_2 was determined to be 3.92 eV. The optical properties of these materials were determined in the UV and EUV range from the REELS analysis and afforded a good agreement with previously reported results for both alloys and PuO_2. The experimentally determined refractive index and extinction coefficient for α-Pu_2O_3 have been

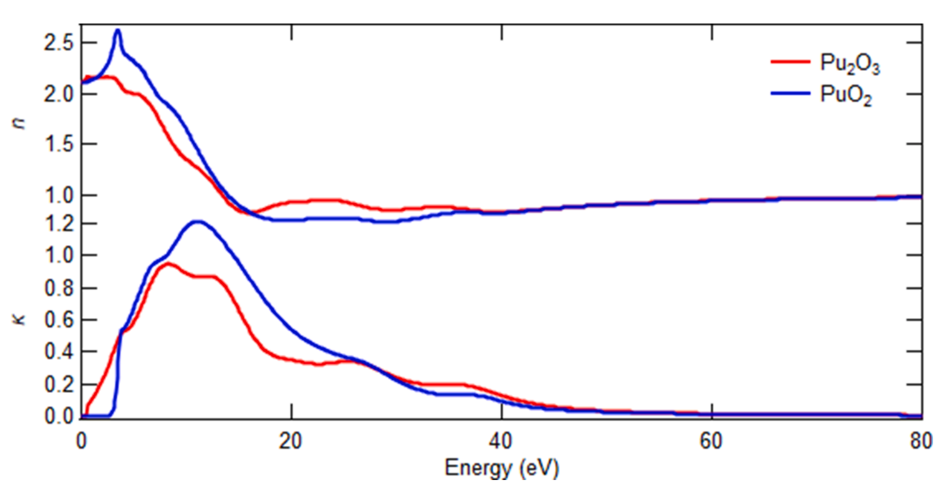

Fig. 7. The optical properties, n and κ for α-Pu_2O_3 and PuO_2 determined from REELS.

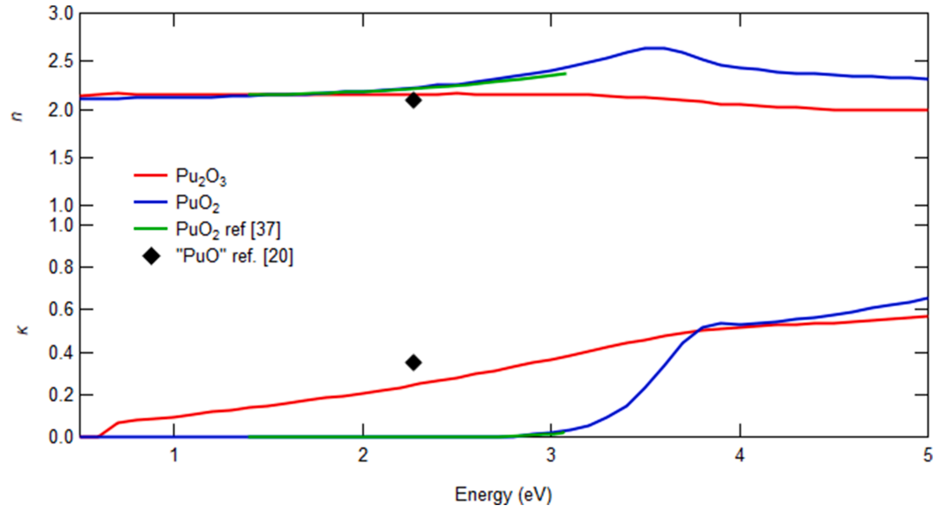

Fig. 8. expanded region of Fig. 7 comparing the optical properties of plutonium oxides determined in this study to the values obtained from ellipsometry.

presented for the first time, and together with the optical properties of plutonium alloys and PuO_2 provide an excellent data set to compare to first-principles calculations. Finally, the optical properties determined here were used to afford an understanding of the oxide composition in the low temperature oxidation study reported by Larson *et al.* and suggest the oxide is predominantly α-Pu_2O_3 within the oxygen pressure and temperatures studied [20].

CRediT authorship contribution statement

P. Roussel: Conceptualization, Methodology, Formal analysis, Investigation, Writing - original draft, Writing - review & editing, Visualization. **K.S. Graham:** Resources. **S.C. Hernandez:** Investigation, Writing - review & editing. **J.J. Joyce:** Investigation, Writing - review & editing. **A.J. Nelson:** Conceptualization, Investigation, Writing - review & editing, Project administration, Supervision. **R. Sykes:** Resources. **T. Venhaus:** Investigation, Project administration, Supervision. **K. White:** Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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